RAVJODH HEER

Platform Engineer | Site Reliability Engineer
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PROFESSIONAL SUMMARY

Platform and reliability engineer with 3+ years operating enterprise-scale Kubernetes infrastructure in production, now supporting R&D backend cloud operations at TP-Link Systems. Three years at AWS on the Containers & IaC team resolving 1,200+ escalated enterprise cases and leading root-cause investigations on multi-team incidents involving etcd, CoreDNS, and CNI networking at 300+ node scale. Strong background in GitOps, Terraform, Helm, Karpenter, and Prometheus/Grafana observability, plus Go and Python automation that turns recurring failure patterns into durable platform improvements.

EXPERIENCE

Cloud Operations Engineer II - R&D Backend | TP-Link Systems Inc. | Irvine, CAOct 2026 - Present

- Support cloud infrastructure operations for R&D backend services, assisting with the deployment, scaling, and day-to-day health of containerized workloads on Kubernetes across development, staging, and production environments.
- Assist with Terraform-based infrastructure provisioning and GitOps-driven release workflows, helping keep environment configuration consistent, reviewable, and version-controlled across AWS, GCP, and Azure.
- Contribute to monitoring, alerting, and incident response for backend platform services, participating in triage and root-cause analysis and documenting findings as runbooks for the operations team.
- Help build operational tooling and automation in Go and Python to reduce manual toil in deployments, configuration changes, and environment management for R&D engineering teams.

Cloud Support Engineer II - Containers & IaC | Amazon Web Services | Remote / San Francisco, CAJun 2023 - May 2026

- Led cross-functional root-cause investigation into an etcd compaction incident on a 300+ node production EKS cluster, coordinating with VPC, RDS, and etcd infrastructure engineering teams to restore cluster health and prevent recurrence.
- Diagnosed a CoreDNS under-scaling incident causing intermittent service resolution failures at scale, tuning autoscaling and resource allocation to stabilize DNS latency cluster-wide.
- Engineered an IPv4 exhaustion remediation playbook using VPC CNI prefix delegation, restoring pod scheduling capacity on large node groups without cluster recreation, VPC redesign, or application downtime.
- Maintained a multi-cluster autoscaling baseline covering Karpenter NodePools, disruption budgets, and Spot fallback topology adopted by 6+ enterprise platform teams; rightsized consolidation policies for a ~35% reduction in Spot idle cost.
- Architected a shared ALB consolidation strategy for multi-tenant EKS using IngressClass group.name segmentation across 7 clusters, eliminating redundant per-service load balancers.
- Built Terraform modules and Helm charts to standardize EKS cluster configuration, IRSA-based IAM access, and observability tooling, and led IaC reviews identifying state drift and IAM privilege escalation paths across multi-account AWS Organizations.
- Resolved 1,200+ high-severity escalated cases spanning HPA, service discovery, node pressure, Linux/systemd failures, VPC networking, and autoscaling, and authored a Python-assisted runbook library adopted org-wide that cut average resolution time by 60%.

Software Engineer | SproutLabs | Santa Cruz, CASep 2022 - Jun 2023

- Built RESTful backend services and CI/CD pipelines with unit, integration, and end-to-end test coverage in a 4-person agile team, establishing deployment reliability practices for production releases.
- Designed a real-time IoT telemetry dashboard integrating farm sensor data through an API-driven architecture, providing device monitoring, analytics, and admin control for end users.

PROJECTS

Kube Cost Manager (KCM) | Founder & Sole DeveloperNov 2024 - Present

- Built a multi-account Kubernetes cost and governance platform on AWS using Terraform, Helm, FastAPI, Lambda, SQS, and EventBridge, with GitOps delivery via Argo CD, drift detection, and OPA/Gatekeeper policy enforcement.
- Implemented Prometheus, Grafana, and CloudWatch observability with dashboards for autoscaler efficiency, node utilization, and cost-per-namespace attribution, letting teams self-serve cost data without platform team involvement.

TECHNICAL SKILLS

Kubernetes & Containers: Kubernetes, EKS, ECS/Fargate, Karpenter, Cluster Autoscaler, Helm, containerd/CRI-O, VPC CNI, CoreDNS, HPA, CRDs, admission webhooks, OPA/Gatekeeper, kubelet/etcd internals

Cloud Platforms: AWS (EC2, Lambda, IAM/IRSA, VPC, Transit Gateway, PrivateLink, Route53, ALB/NLB, SQS, EventBridge), GCP, Azure

IaC, GitOps & CI/CD: Terraform, GitOps, Argo CD, CloudFormation, AWS CDK, GitHub Actions, GitLab CI, Jenkins

Observability & Reliability: Prometheus, Grafana, CloudWatch, OpenTelemetry, Datadog, ELK Stack, SLO/SLI alerting, incident response, RCA, post-mortems, error budgets

Linux & Networking: Linux, systemd, cgroups, journald, TCP/IP, DNS, TLS, HTTP, routing, capacity planning

Languages & Security: Go, Python, Bash, TypeScript, SQL; IAM policy design, RBAC, Pod Security Standards, secrets management

EDUCATION & CERTIFICATIONS

B.S. Computer Science | University of California, Santa CruzJun 2023

AWS Certified Solutions Architect - Associate | AWS Certified Cloud Practitioner